



sunflower



Lotus leaf



Bird wings



Frog skin

BIO SMART ROOF

Presenters

S.A.M.Prabhani (E/25/300)
G.K.O.Thilasha (E/25/398)
H.W.A.Thamarangi (E/25/385)
S.A.Sathruwani (E/25/359)

CONTENT

❖ Sun Flower

- Solar tracking
- Rotation mechanics
- Engineering application

❖ Bird Wings

- Adaptive folding roof
- Engineering application

❖ Frog Skin

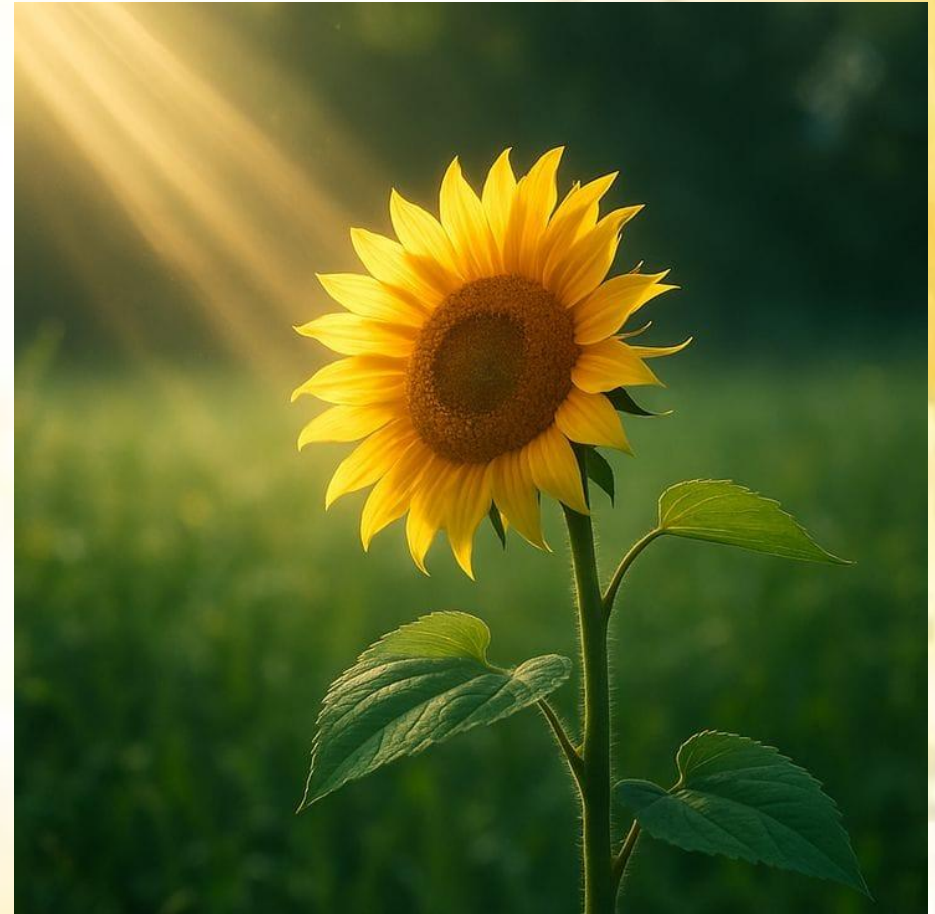
- Moisture management
- Engineering application

❖ Lotus Leaf

- Self Cleaning System
- Engineering Application

SUNFLOWER (SOLAR TRACKING)

- ☐ Natural response to sunlight
- ☐ Movement towards the sun
- ☐ Direction adjustment according to sunlight
- ☐ Efficient sunlight absorption



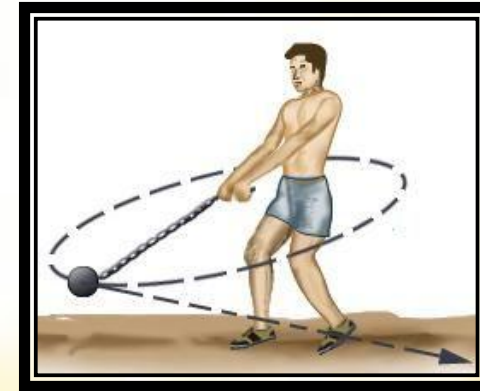
SOLAR PANEL ROTATION MECHANICS



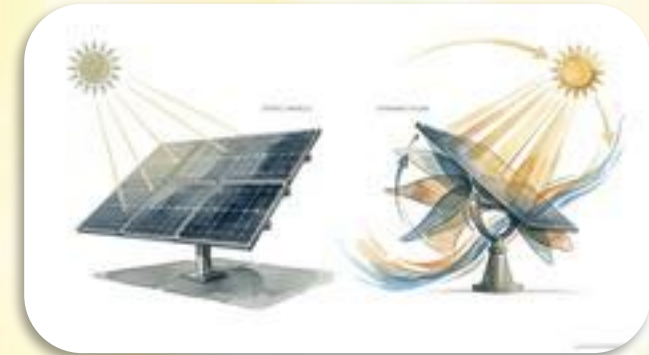
Vertical axis(Azimuth rotation)



Horizontal axis(elevation rotation)



- ✓ Servo motors or geared motors
- ✓ Rotating base
- ✓ Central shaft and bearings



ENGINEERING APPLICATION

- ☐ Light sensors
- ☐ Sunlight direction detection
- ☐ Microcontroller-based control
- ☐ Automatic motor movement
- ☐ Adjustable solar panels
- ☐ Improved solar energy collection

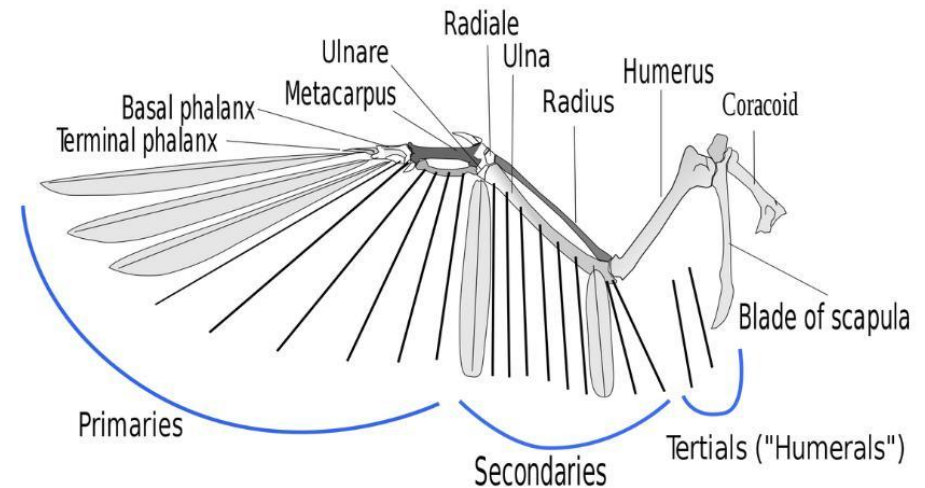


Flow Chart – Solar Tracking System



BIRD WINGS (ADAPTIVE FOLDING ROOF)

- ☐ Fold
- ☐ Unfold
- ☐ Flexible movement
- ☐ Environmental response



ENGINEERING APPLICATION

- ☐ Rain sensor
- ☐ Heavy rain detection
- ☐ Automatic folding
- ☐ Bird-wing-inspired mechanism
- ☐ Rain-based movement
- ☐ Automatic unfolding



Flow Chart – Adaptive Folding Roof System



FROG SKIN(MOISTURE MANAGEMENT)

- ❑ Thin and permeable skin
- ❑ Water adsorption through skin
- ❑ Mucus glands for moisture retention
- ❑ Skin protection through moisture



ENGINEERING APPLICATION

- ☐ Moisture level detection
- ☐ Sensor data transmission
- ☐ Microcontroller monitoring
- ☐ Excess moisture identification
- ☐ Automatic ventilation
- ☐ Smart drainage activation



LOTUS LEAF(SELF-CLEANING SYSTEM)

- ☐ Water-repellent surface
- ☐ Rolling water droplets
- ☐ Dust and dirt removal
- ☐ Natural self-cleaning effect
- ☐ Lotus effect phenomenon



ENGINEERING APPLICATION

- ☐ Water-repellent roof coating
- ☐ Rainwater droplet formation
- ☐ Dust and dirt removal
- ☐ Natural roof cleaning
- ☐ Reduced manual maintenance

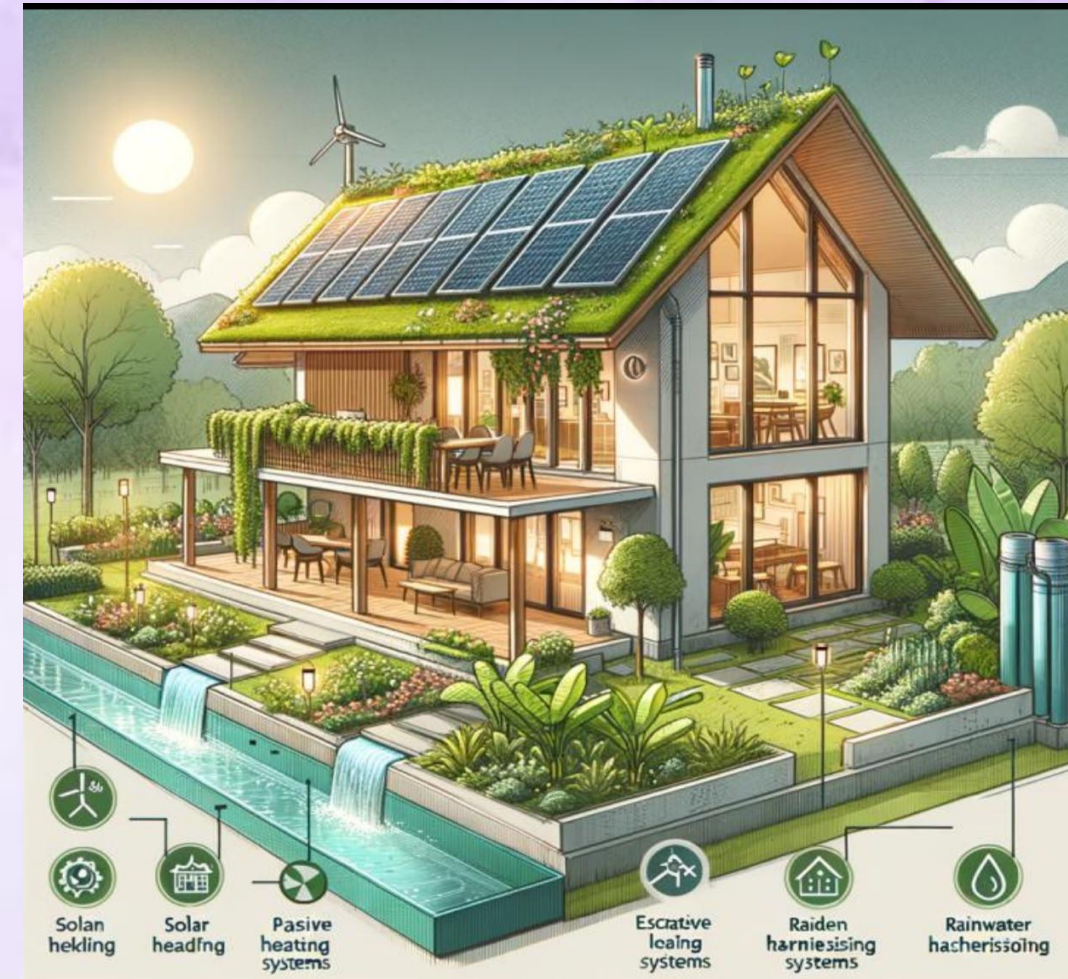


FINAL CONCEPT – NatureSync

NatureSync – Self-Adaptive Eco Roof

Inspired by Nature, Designed for the Future

Natural Inspiration	Engineering Application
 Sunflower	Solar Tracking
 Frog Skin	Moisture management
 Lotus Leaf	Self-Cleaning Surface
 Bird Wings	Adaptive Folding Roof



RESOURCES

- <https://chatgpt.com>
- <https://www.pinterest.com>

Q & A

The image features a light blue background with soft, white, cloud-like patterns. In the corners, there are decorative illustrations of blue roses and green foliage. The top right corner shows a large blue rose and a smaller one with green leaves and stems. The bottom left corner also features blue roses and greenery. The text 'Thank You!' is centered in a large, bold, blue serif font.

Thank You!